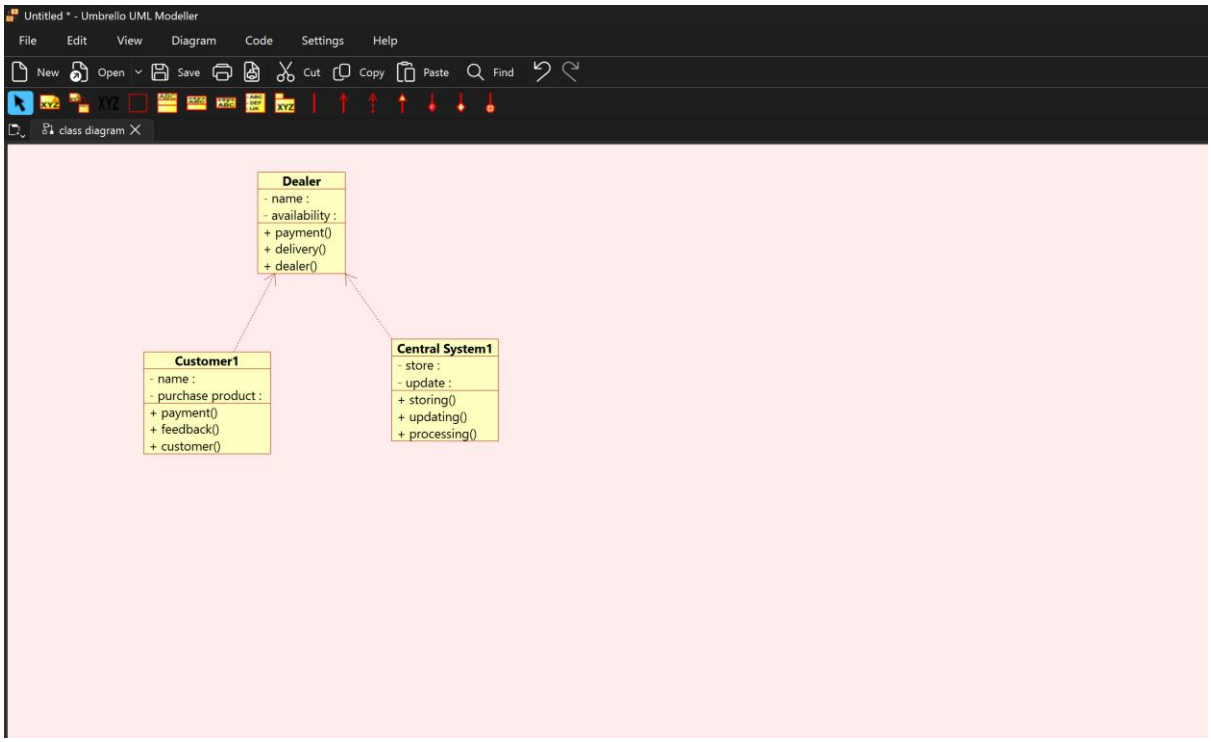
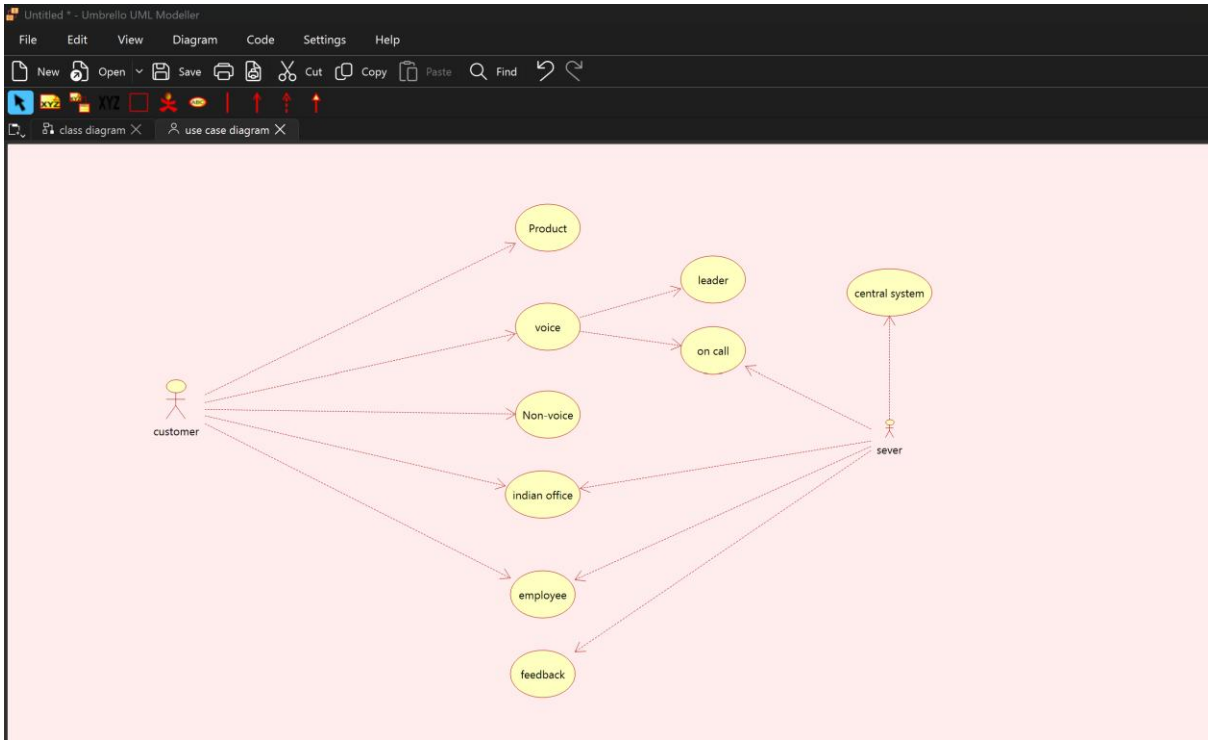
Ex No: 13 – BPO MANANGEMENT SYSTEM

NAME: HEMALAKSHMI H (192571049)

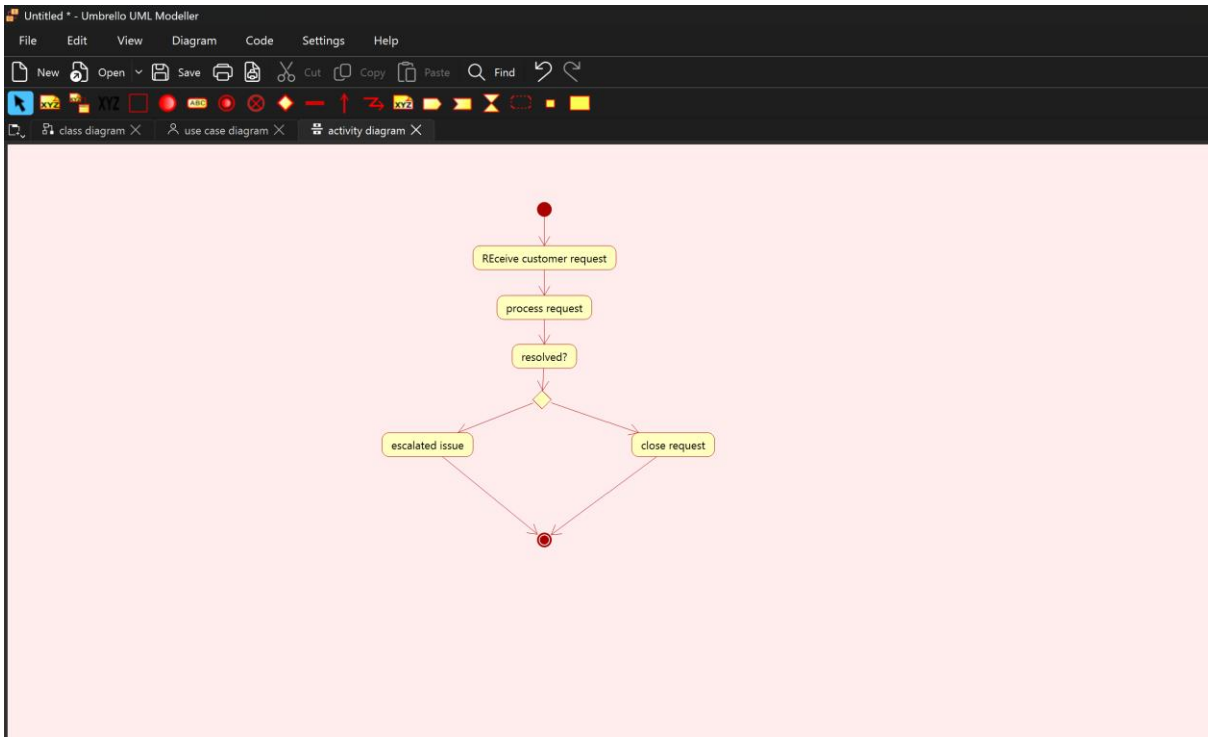
CLASS DIAGRAM:



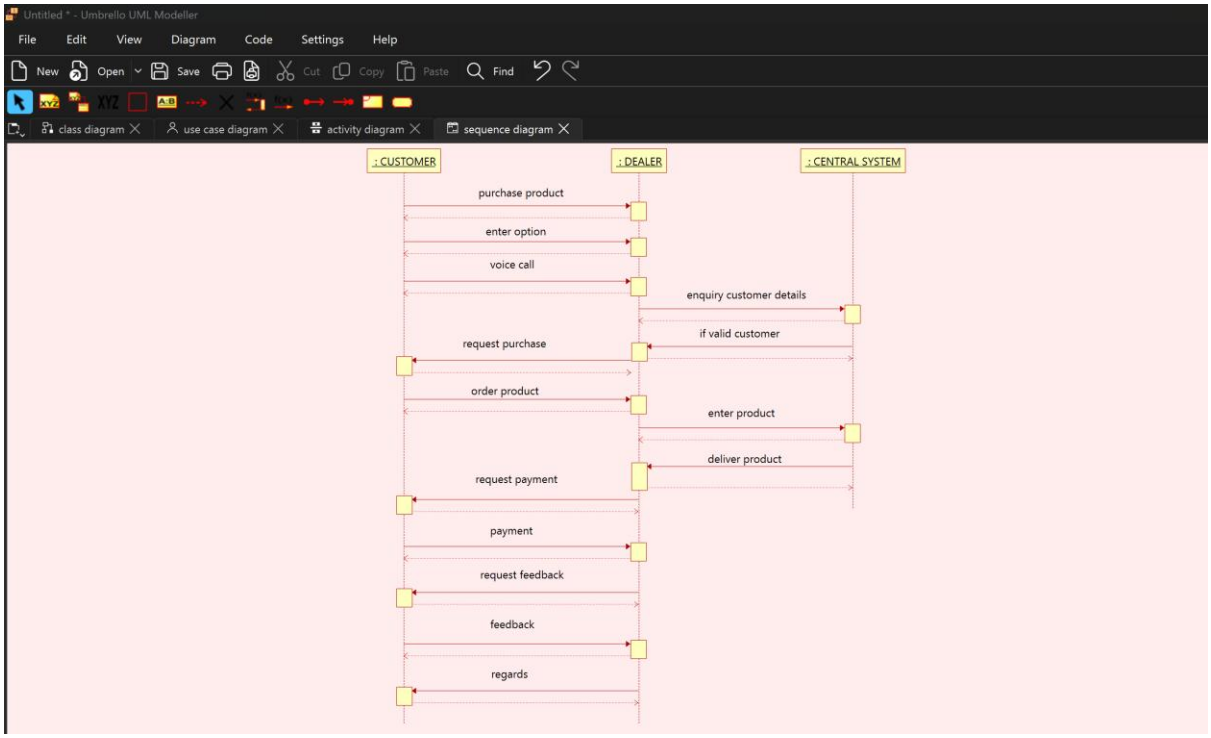
USE CASE DIAGRAM:



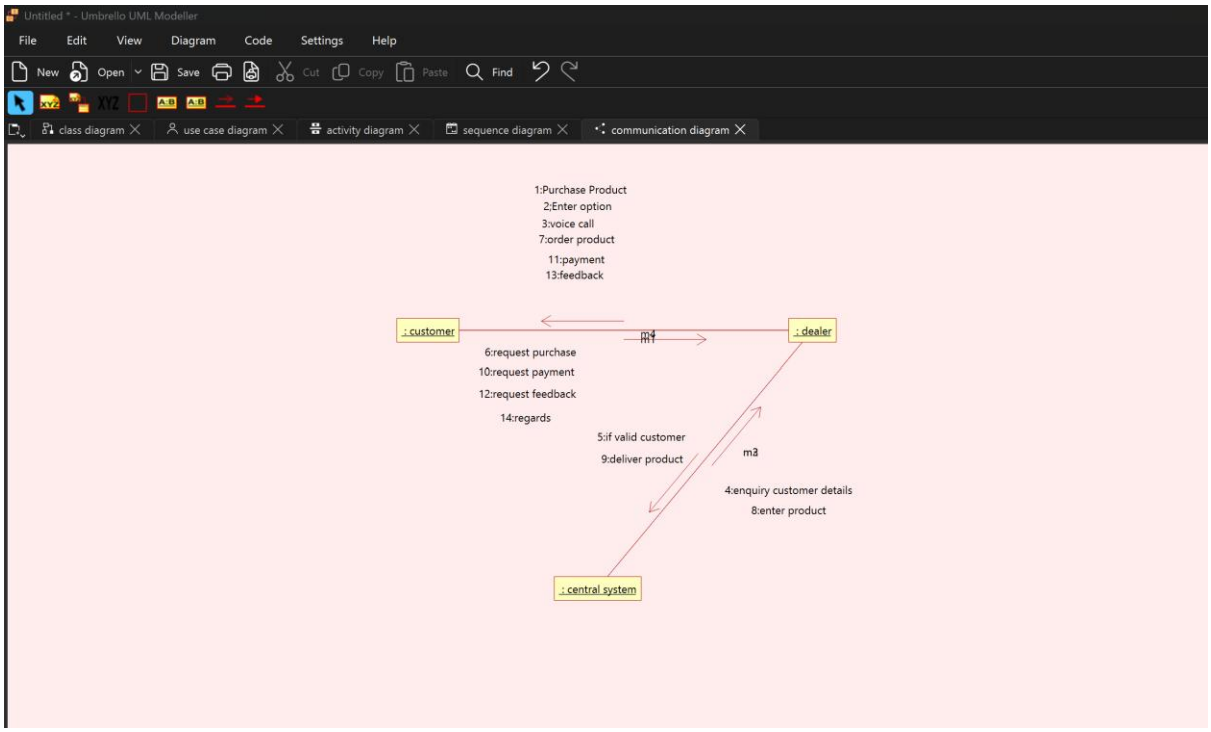
ACTIVITY DIAGRAM:



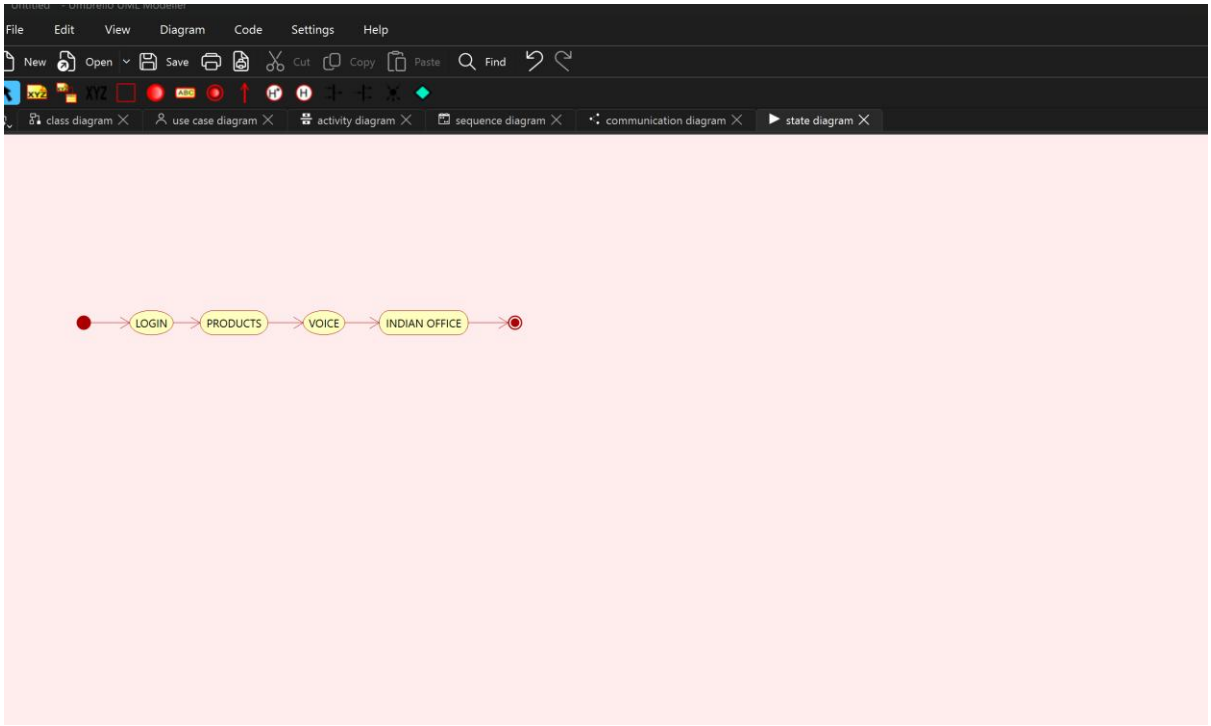
SEQUENCE DIAGRAM:



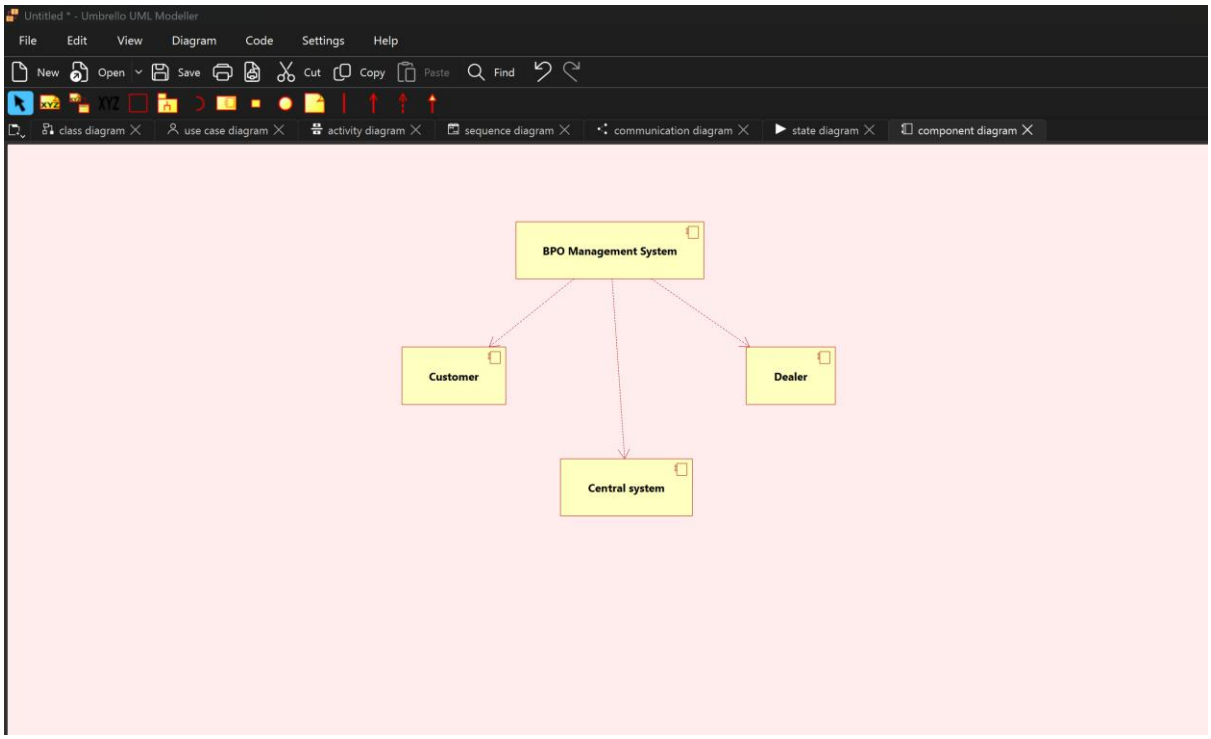
COLLABORATION DIAGRAM:



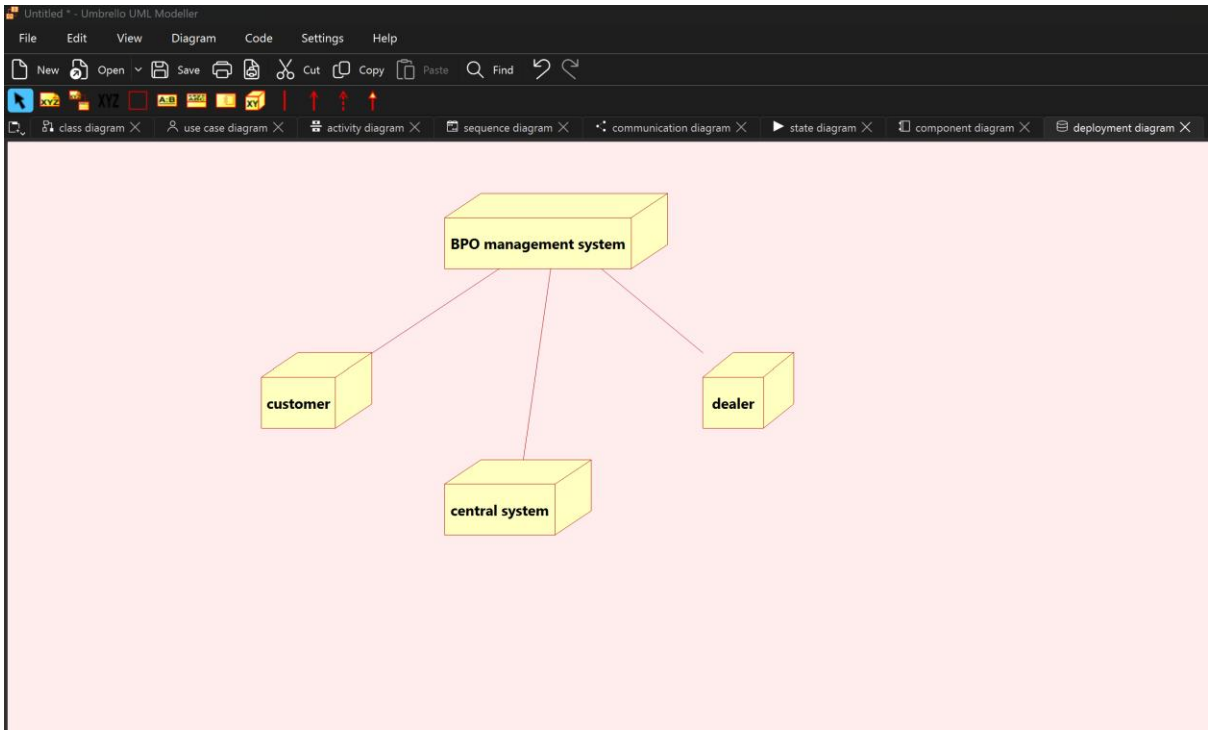
STATE CHART DIAGRAM:



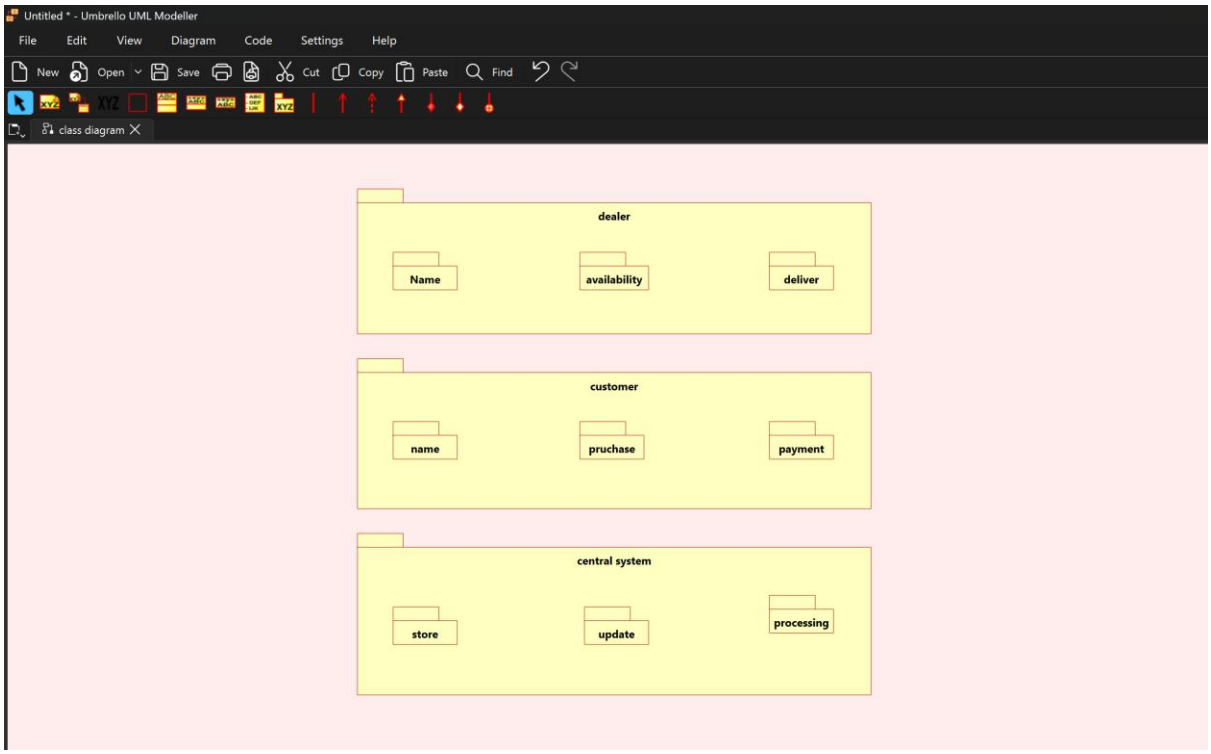
COMPONENT DIAGRAM:



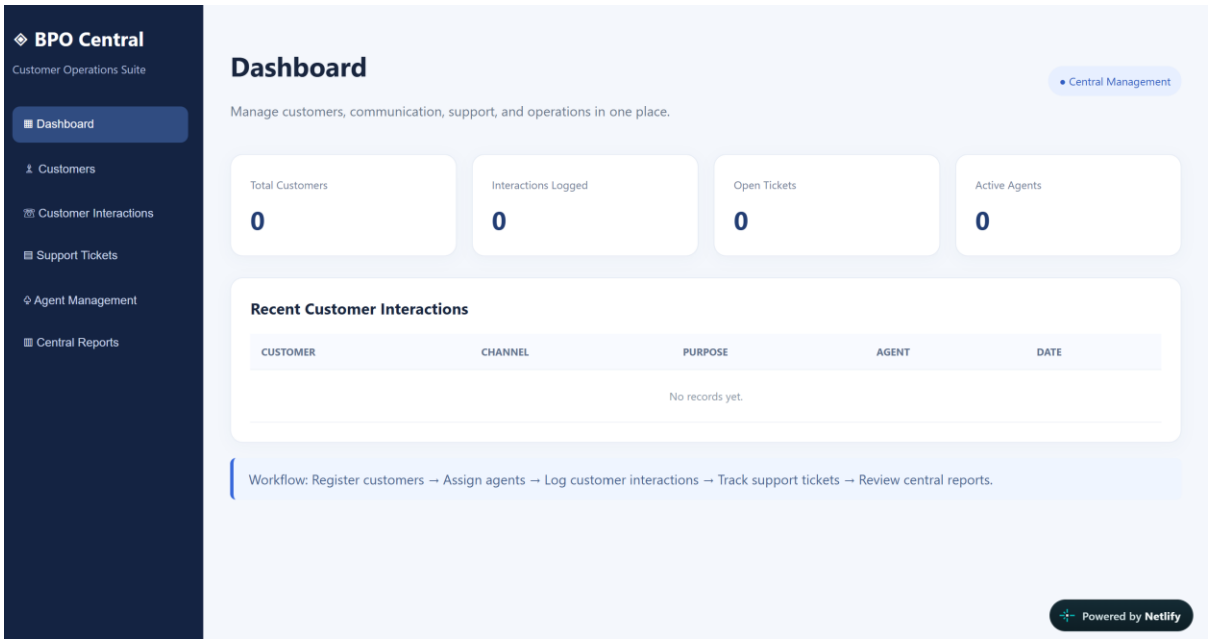
DEPLOYMENT DIAGRAM:

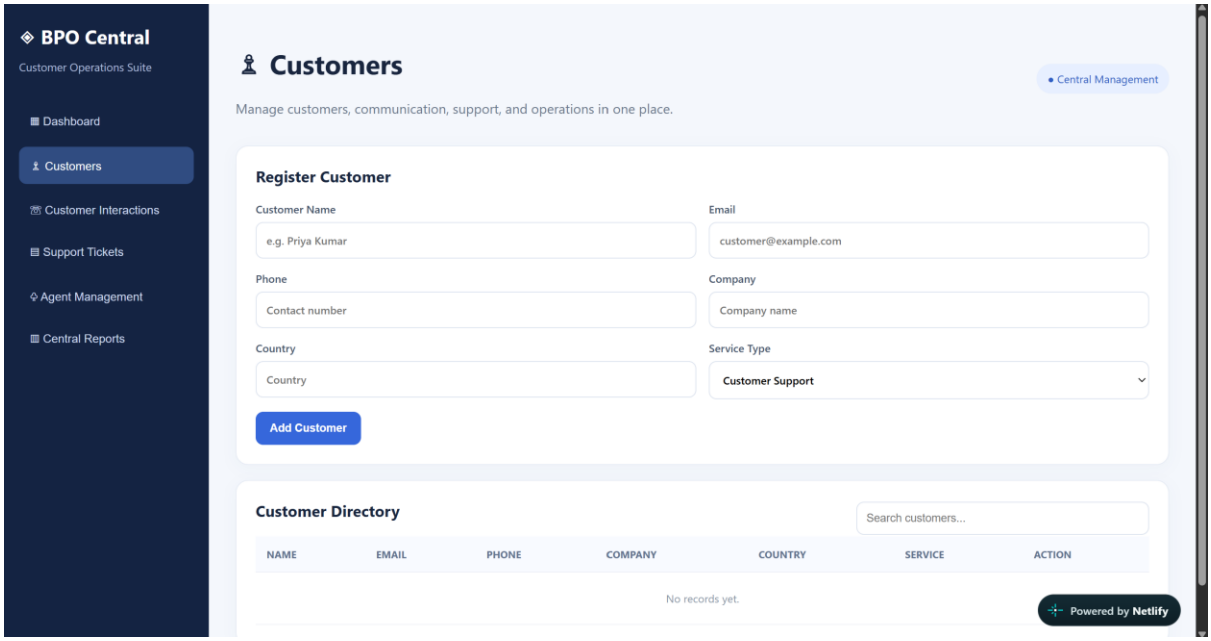


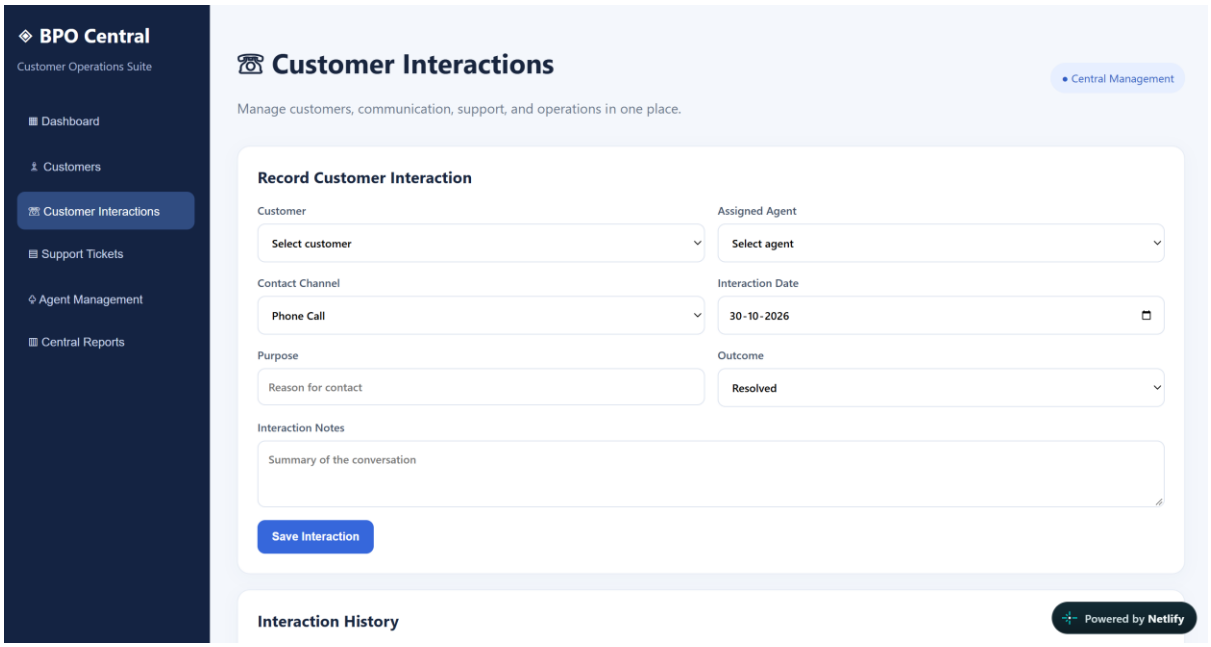
PACKAGE DIAGRAM:

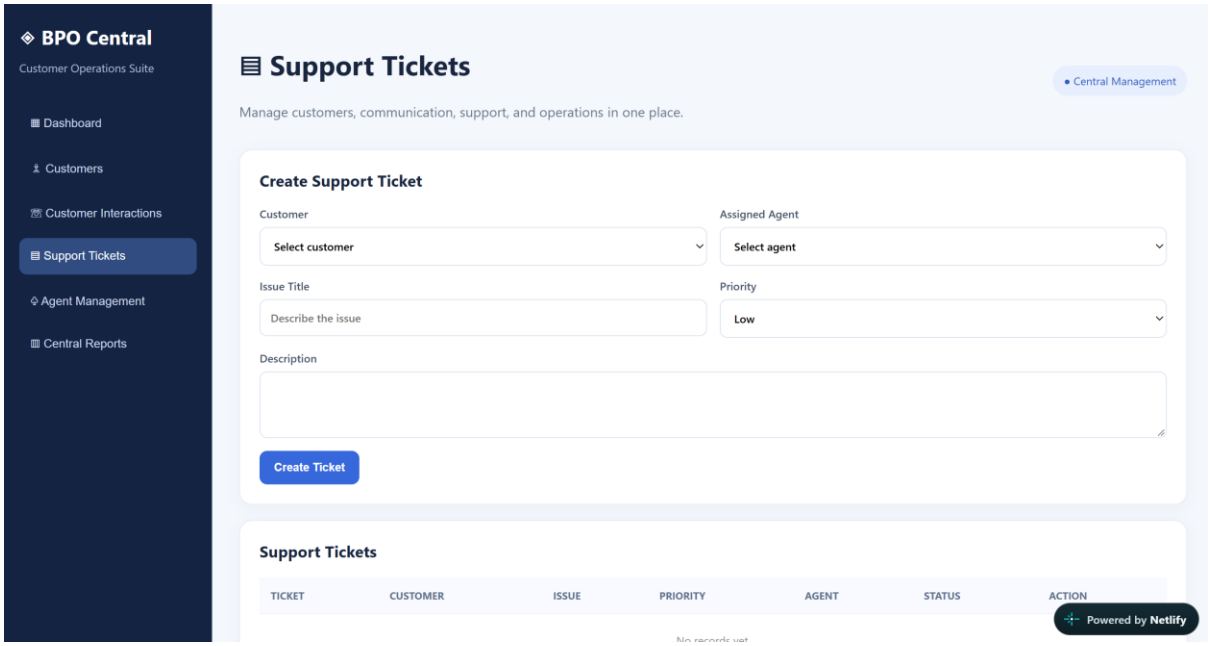


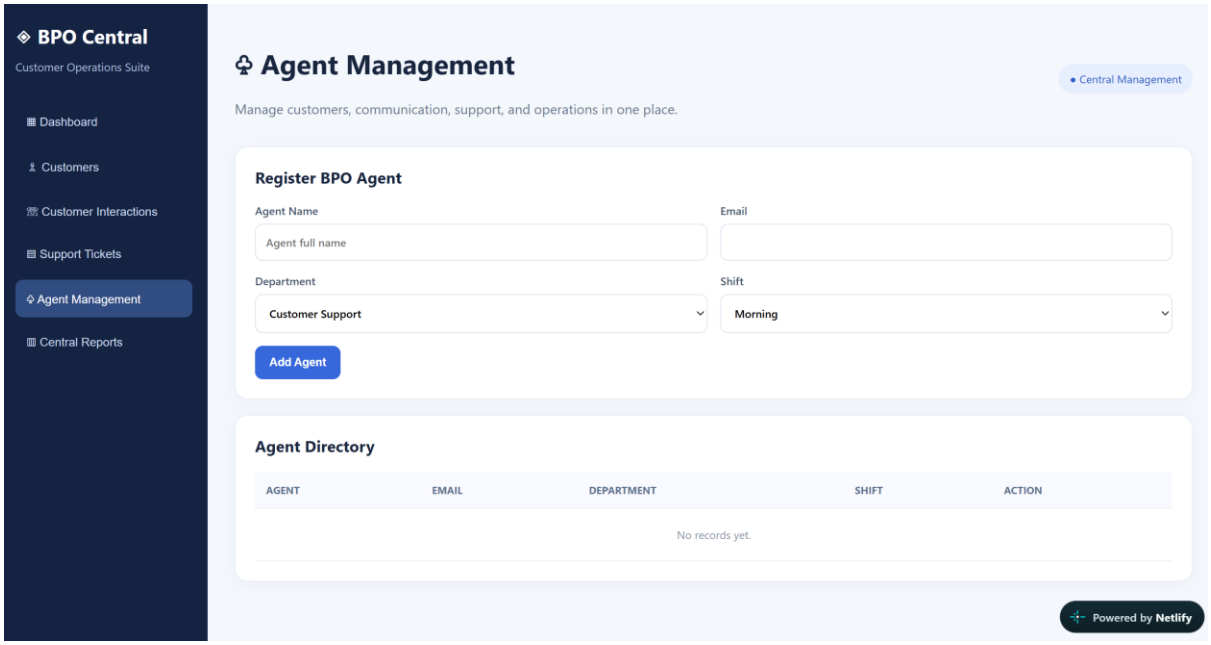
WEBPAGE FOR BPO MANANGEMENT SYSTEM

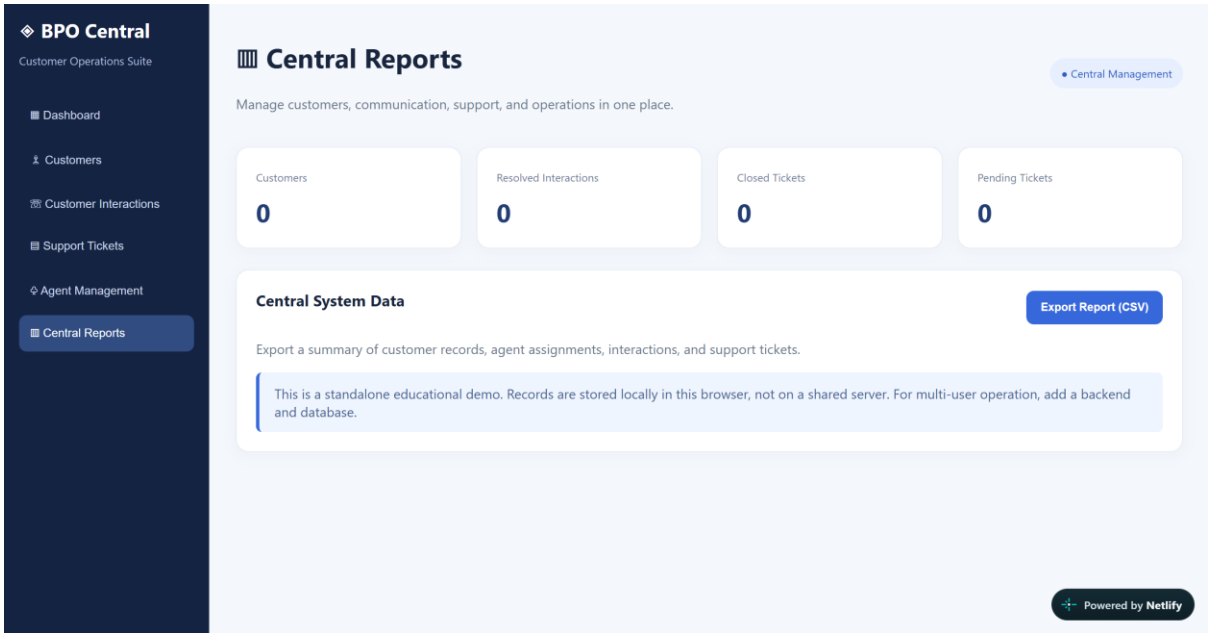



BPO Central
Customer Operations Suite
Dashboard
Customers
Customer Interactions
Support Tickets
Agent Management
Central Reports
Customers
Central Management
Manage customers, communication, support, and operations in one place.
Register Customer
Customer Name
e.g. Priya Kumar
Email
customer@example.com
Phone
Contact number
Company
Company name
Country
Country
Service Type
Customer Support
Add Customer
Customer Directory
Search customers...
NAME
EMAIL
PHONE
COMPANY
COUNTRY
SERVICE
ACTION
No records yet.
Powered by Netlify

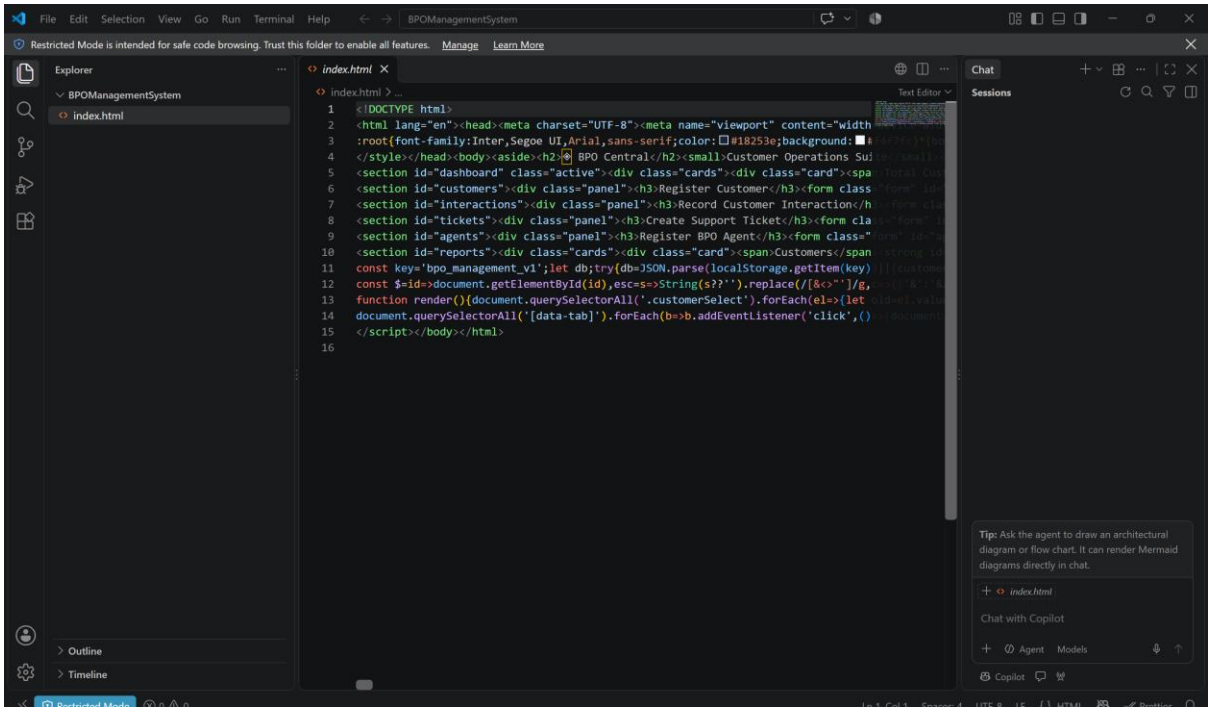

BPO Central
Customer Operations Suite
Dashboard
Customers
Customer Interactions
Support Tickets
Agent Management
Central Reports
Customer Interactions
Central Management
Manage customers, communication, support, and operations in one place.
Record Customer Interaction
Customer
Select customer
Assigned Agent
Select agent
Contact Channel
Phone Call
Interaction Date
30-10-2026
Purpose
Reason for contact
Outcome
Resolved
Interaction Notes
Summary of the conversation
Save Interaction
Interaction History
Powered by Netlify


BPO Central
Customer Operations Suite
Dashboard
Customers
Customer Interactions
Support Tickets
Agent Management
Central Reports
Support Tickets
Central Management
Manage customers, communication, support, and operations in one place.
Create Support Ticket
Customer
Select customer
Assigned Agent
Select agent
Issue Title
Describe the issue
Priority
Low
Description
Create Ticket
Support Tickets
TICKET
CUSTOMER
ISSUE
PRIORITY
AGENT
STATUS
ACTION
Powered by Netlify

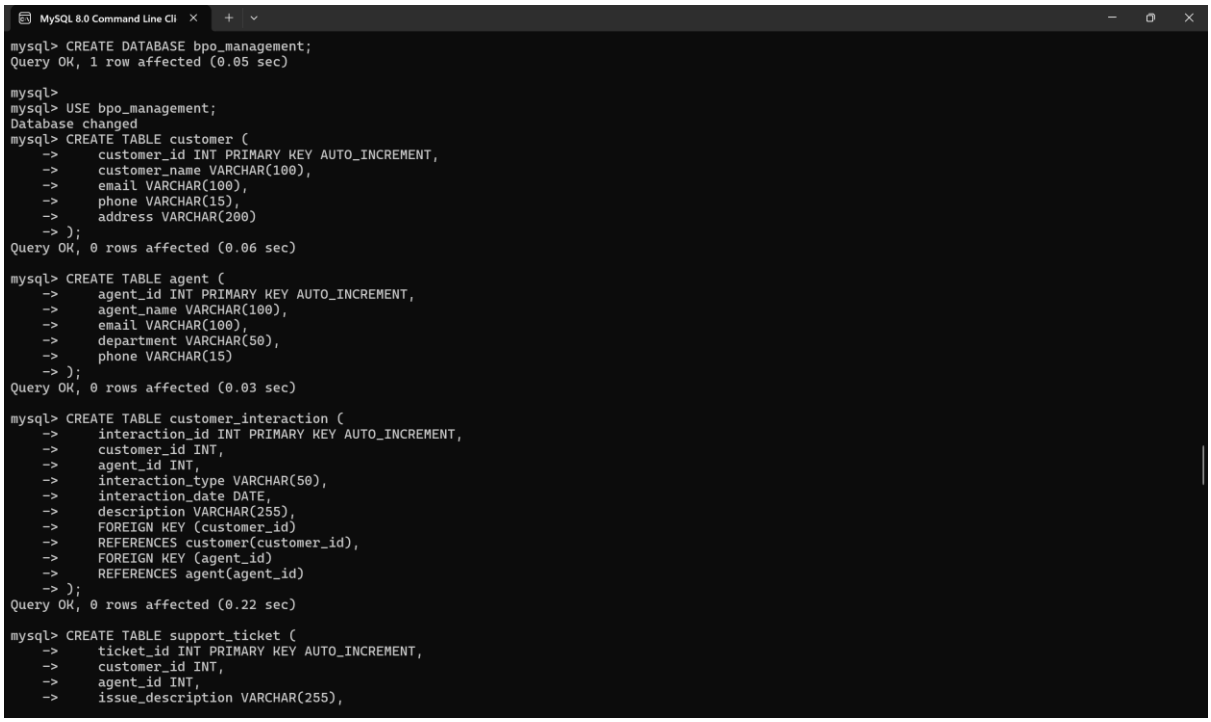

BPO Central
Customer Operations Suite
Dashboard
Customers
Customer Interactions
Support Tickets
Agent Management
Central Reports
Agent Management
Central Management
Manage customers, communication, support, and operations in one place.
Register BPO Agent
Agent Name
Agent full name
Email
Department
Customer Support
Shift
Morning
Add Agent
Agent Directory
AGENT
EMAIL
DEPARTMENT
SHIFT
ACTION
No records yet.
Powered by Netlify


BPO Central
Customer Operations Suite
Dashboard
Customers
Customer Interactions
Support Tickets
Agent Management
Central Reports
Central Reports
Central Management
Manage customers, communication, support, and operations in one place.
Customers
0
Resolved Interactions
0
Closed Tickets
0
Pending Tickets
0
Central System Data
Export Report (CSV)
Export a summary of customer records, agent assignments, interactions, and support tickets.
This is a standalone educational demo. Records are stored locally in this browser, not on a shared server. For multi-user operation, add a backend and database.
Powered by Netlify

HTML CODE:


Restricted Mode is intended for safe code browsing. Trust this folder to enable all features.
Manage
Learn More
Explorer
BPOManagementSystem
index.html
Outline
Timeline
Chat
Sessions
Tip: Ask the agent to draw an architectural diagram or flow chart. It can render Mermaid diagrams directly in chat.
Chat with Copilot
Agent
Models
Copilot

MYSQL DATABASE:



```
MySQL 8.0 Command Line Cli

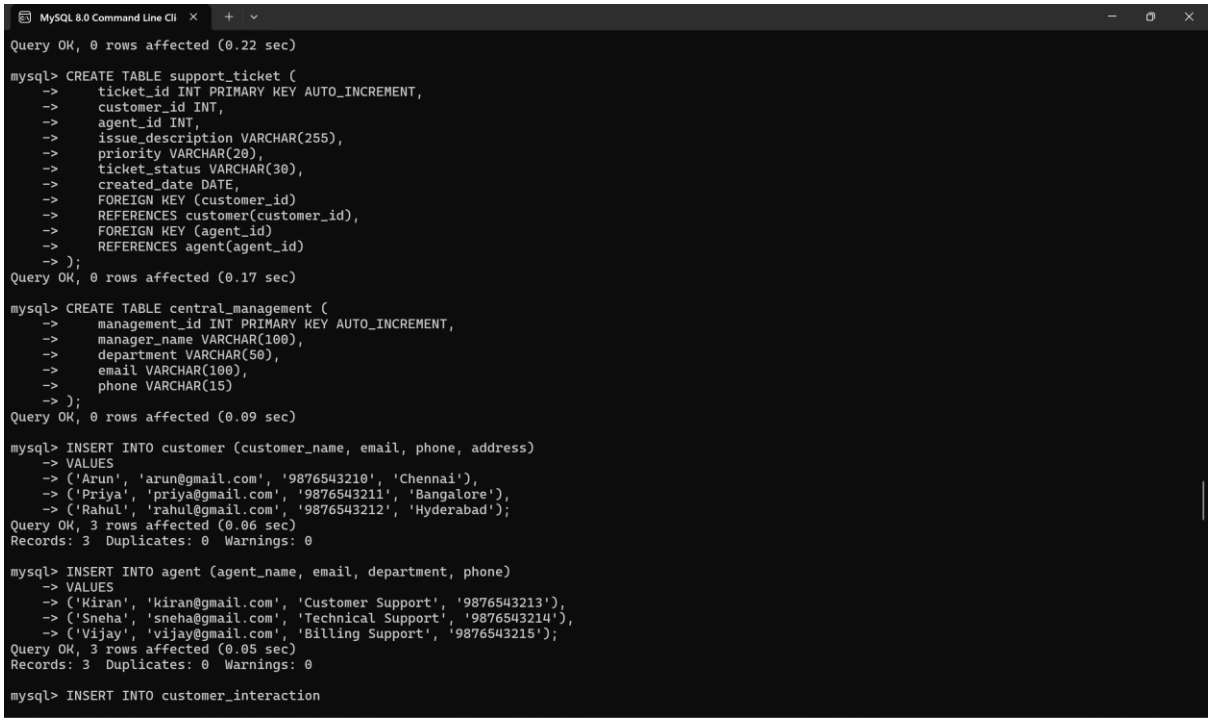
mysql> CREATE DATABASE bpo_management;
Query OK, 1 row affected (0.05 sec)

mysql>
mysql> USE bpo_management;
Database changed
mysql> CREATE TABLE customer (
    ->     customer_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     customer_name VARCHAR(100),
    ->     email VARCHAR(100),
    ->     phone VARCHAR(15),
    ->     address VARCHAR(200)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> CREATE TABLE agent (
    ->     agent_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     agent_name VARCHAR(100),
    ->     email VARCHAR(100),
    ->     department VARCHAR(50),
    ->     phone VARCHAR(15)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE customer_interaction (
    ->     interaction_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     customer_id INT,
    ->     agent_id INT,
    ->     interaction_type VARCHAR(50),
    ->     interaction_date DATE,
    ->     description VARCHAR(255),
    ->     FOREIGN KEY (customer_id)
    ->     REFERENCES customer(customer_id),
    ->     FOREIGN KEY (agent_id)
    ->     REFERENCES agent(agent_id)
    -> );
Query OK, 0 rows affected (0.22 sec)

mysql> CREATE TABLE support_ticket (
    ->     ticket_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     customer_id INT,
    ->     agent_id INT,
    ->     issue_description VARCHAR(255),
```



```
MySQL 8.0 Command Line Cli

Query OK, 0 rows affected (0.22 sec)

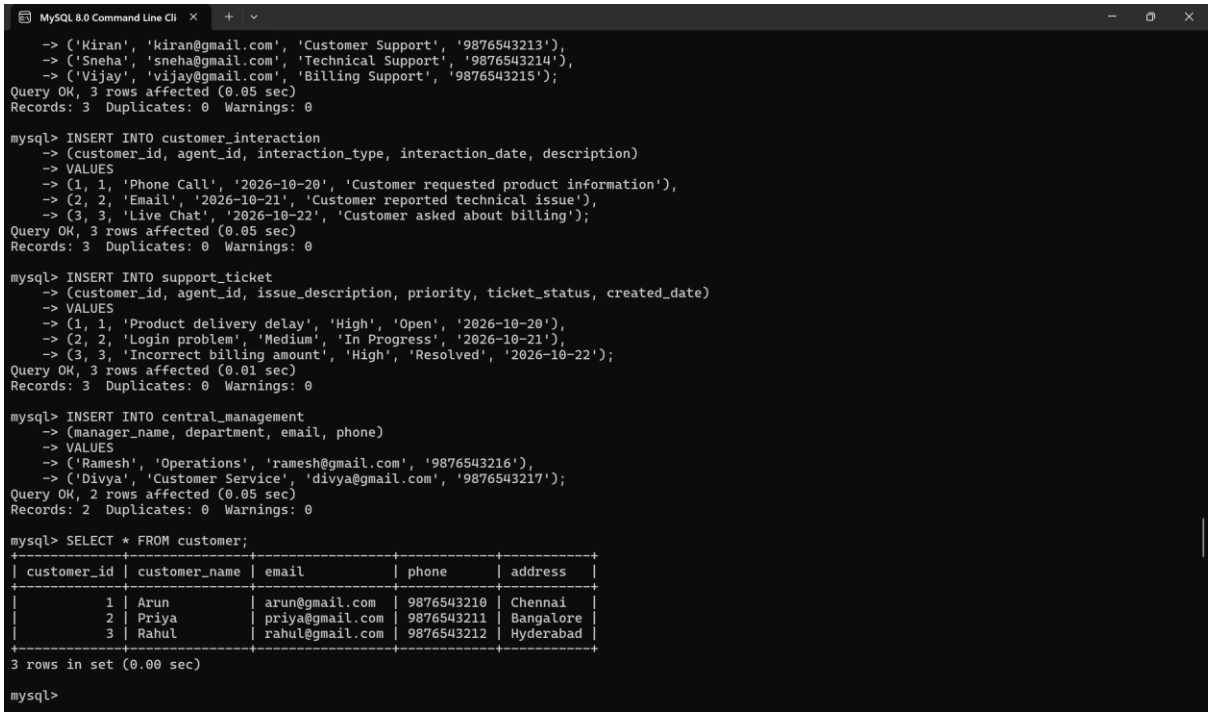
mysql> CREATE TABLE support_ticket (
    ->     ticket_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     customer_id INT,
    ->     agent_id INT,
    ->     issue_description VARCHAR(255),
    ->     priority VARCHAR(20),
    ->     ticket_status VARCHAR(30),
    ->     created_date DATE,
    ->     FOREIGN KEY (customer_id)
    ->     REFERENCES customer(customer_id),
    ->     FOREIGN KEY (agent_id)
    ->     REFERENCES agent(agent_id)
    -> );
Query OK, 0 rows affected (0.17 sec)

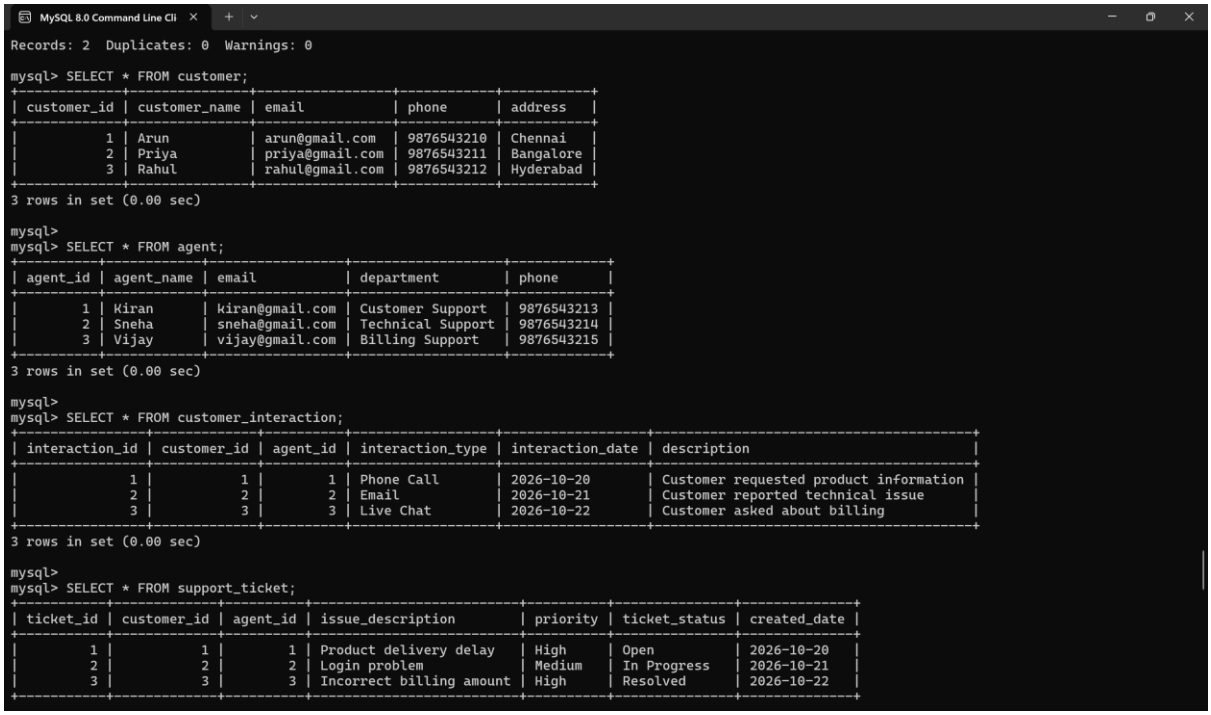
mysql> CREATE TABLE central_management (
    ->     management_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     manager_name VARCHAR(100),
    ->     department VARCHAR(50),
    ->     email VARCHAR(100),
    ->     phone VARCHAR(15)
    -> );
Query OK, 0 rows affected (0.09 sec)

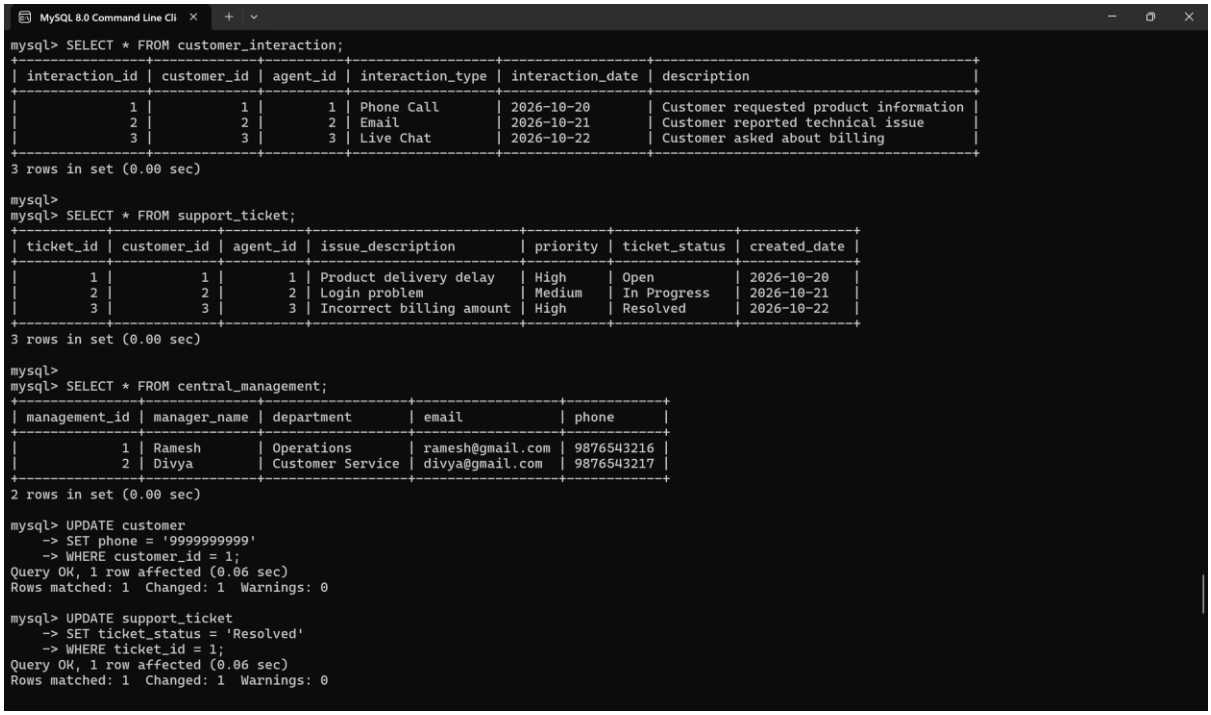
mysql> INSERT INTO customer (customer_name, email, phone, address)
    -> VALUES
    -> ('Arun', 'arun@gmail.com', '9876543210', 'Chennai'),
    -> ('Priya', 'priya@gmail.com', '9876543211', 'Bangalore'),
    -> ('Rahul', 'rahul@gmail.com', '9876543212', 'Hyderabad');
Query OK, 3 rows affected (0.06 sec)
Records: 3  Duplicates: 0  Warnings: 0

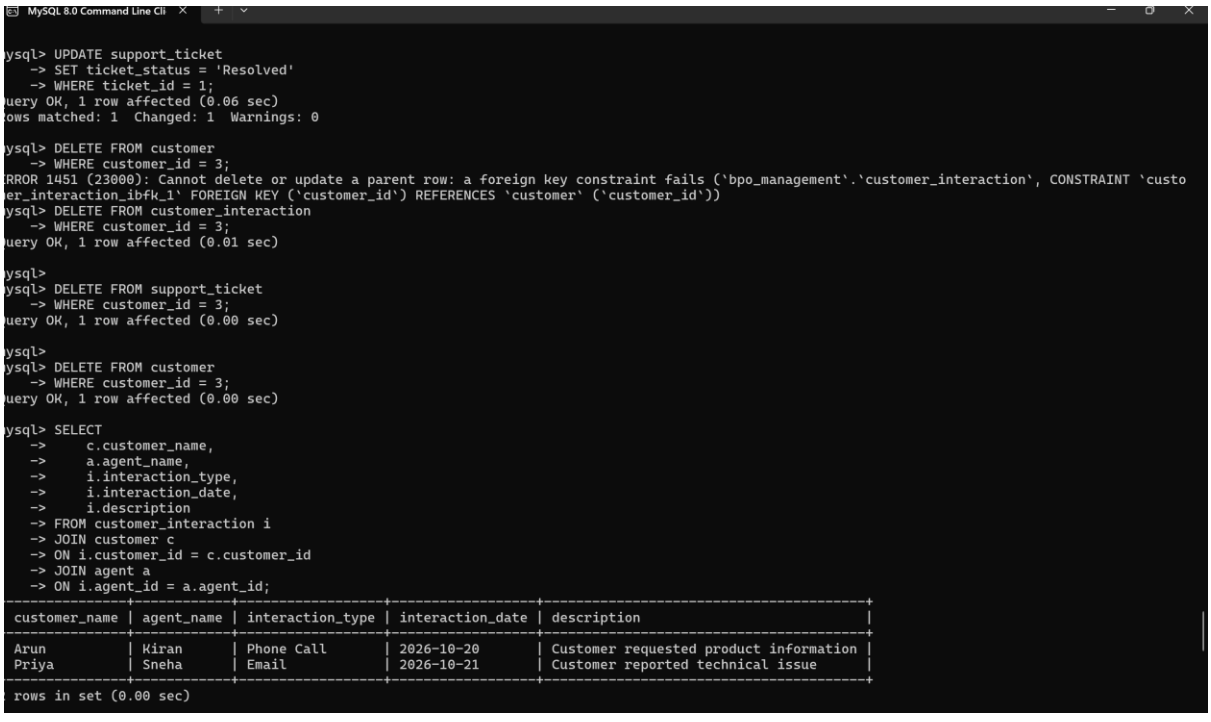
mysql> INSERT INTO agent (agent_name, email, department, phone)
    -> VALUES
    -> ('Kiran', 'kiran@gmail.com', 'Customer Support', '9876543213'),
    -> ('Sneha', 'sneha@gmail.com', 'Technical Support', '9876543214'),
    -> ('Vijay', 'vijay@gmail.com', 'Billing Support', '9876543215');
Query OK, 3 rows affected (0.05 sec)
Records: 3  Duplicates: 0  Warnings: 0

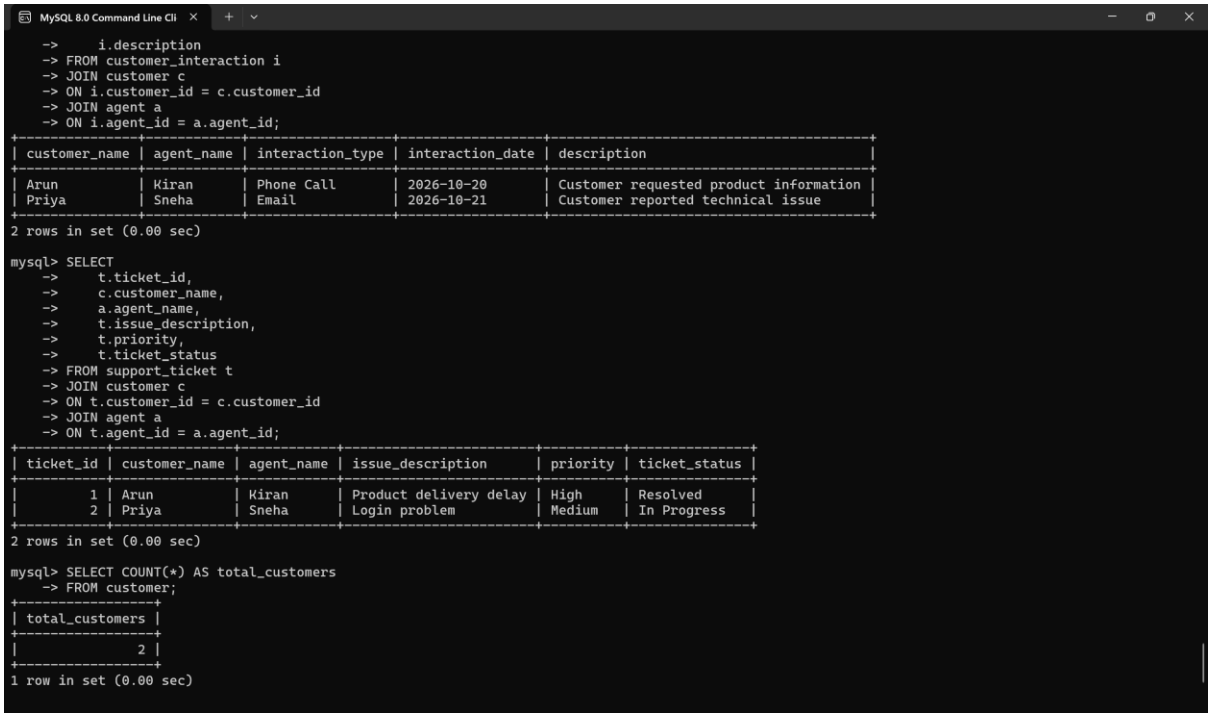
mysql> INSERT INTO customer_interaction
```

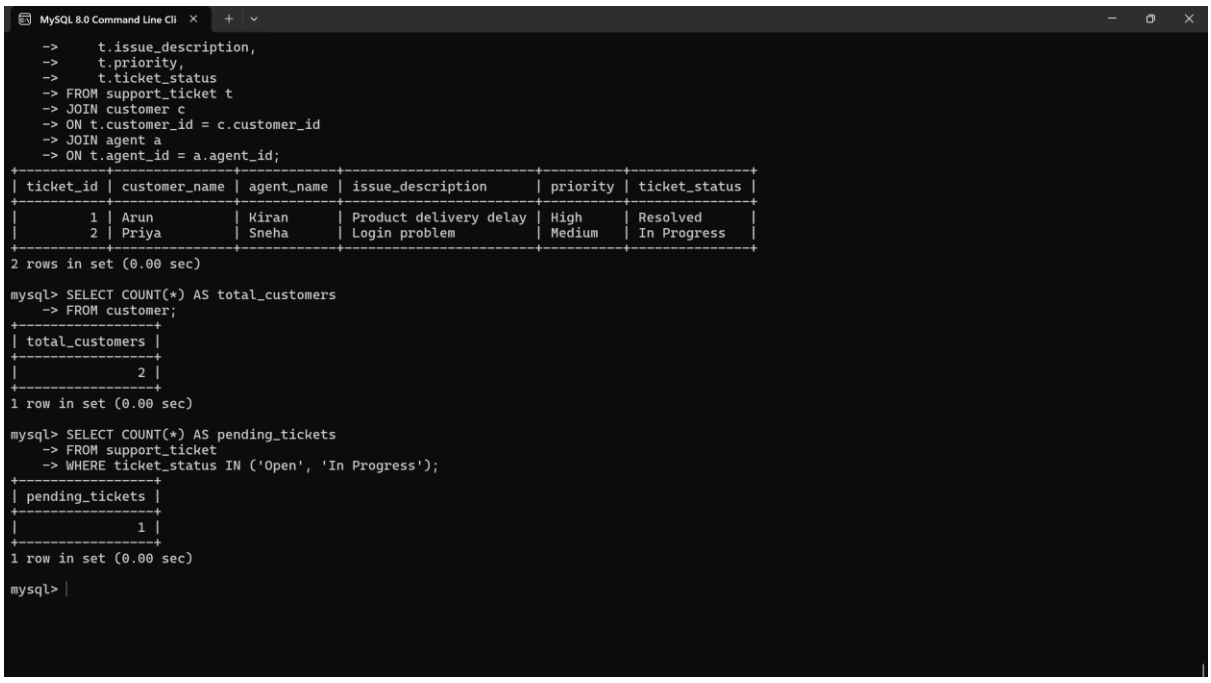

MySQL 8.0 Command Line Cli
-> ('Kiran', 'kiran@gmail.com', 'Customer Support', '9876543213'),
-> ('Sneha', 'sneha@gmail.com', 'Technical Support', '9876543214'),
-> ('Vijay', 'vijay@gmail.com', 'Billing Support', '9876543215');
Query OK, 3 rows affected (0.05 sec)
Records: 3 Duplicates: 0 Warnings: 0
mysql> INSERT INTO customer_interaction
-> (customer_id, agent_id, interaction_type, interaction_date, description)
-> VALUES
-> (1, 1, 'Phone Call', '2026-10-20', 'Customer requested product information'),
-> (2, 2, 'Email', '2026-10-21', 'Customer reported technical issue'),
-> (3, 3, 'Live Chat', '2026-10-22', 'Customer asked about billing');
Query OK, 3 rows affected (0.05 sec)
Records: 3 Duplicates: 0 Warnings: 0
mysql> INSERT INTO support_ticket
-> (customer_id, agent_id, issue_description, priority, ticket_status, created_date)
-> VALUES
-> (1, 1, 'Product delivery delay', 'High', 'Open', '2026-10-20'),
-> (2, 2, 'Login problem', 'Medium', 'In Progress', '2026-10-21'),
-> (3, 3, 'Incorrect billing amount', 'High', 'Resolved', '2026-10-22');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0
mysql> INSERT INTO central_management
-> (manager_name, department, email, phone)
-> VALUES
-> ('Ramesh', 'Operations', 'ramesh@gmail.com', '9876543216'),
-> ('Divya', 'Customer Service', 'divya@gmail.com', '9876543217');
Query OK, 2 rows affected (0.05 sec)
Records: 2 Duplicates: 0 Warnings: 0
mysql> SELECT * FROM customer;
+-------------+---------------+-----------------+------------+-----------+
| customer_id | customer_name | email | phone | address |
+-------------+---------------+-----------------+------------+-----------+
1	Arun	arun@gmail.com	9876543210	Chennai
2	Priya	priya@gmail.com	9876543211	Bangalore
3	Rahul	rahul@gmail.com	9876543212	Hyderabad
+-------------+---------------+-----------------+------------+-----------+
3 rows in set (0.00 sec)
mysql>


MySQL 8.0 Command Line Cli
Records: 2 Duplicates: 0 Warnings: 0
mysql> SELECT * FROM customer;
+-------------+---------------+-----------------+------------+-----------+
| customer_id | customer_name | email | phone | address |
+-------------+---------------+-----------------+------------+-----------+
1	Arun	arun@gmail.com	9876543210	Chennai
2	Priya	priya@gmail.com	9876543211	Bangalore
3	Rahul	rahul@gmail.com	9876543212	Hyderabad
+-------------+---------------+-----------------+------------+-----------+
3 rows in set (0.00 sec)
mysql>
mysql> SELECT * FROM agent;
+----------+------------+-----------------+-------------------+------------+
| agent_id | agent_name | email | department | phone |
+----------+------------+-----------------+-------------------+------------+
1	Kiran	kiran@gmail.com	Customer Support	9876543213
2	Sneha	sneha@gmail.com	Technical Support	9876543214
3	Vijay	vijay@gmail.com	Billing Support	9876543215
+----------+------------+-----------------+-------------------+------------+
3 rows in set (0.00 sec)
mysql>
mysql> SELECT * FROM customer_interaction;
+----------------+-------------+----------+------------------+------------------+---------------------------------------+
| interaction_id | customer_id | agent_id | interaction_type | interaction_date | description |
+----------------+-------------+----------+------------------+------------------+---------------------------------------+
1	1	1	Phone Call	2026-10-20	Customer requested product information
2	2	2	Email	2026-10-21	Customer reported technical issue
3	3	3	Live Chat	2026-10-22	Customer asked about billing
+----------------+-------------+----------+------------------+------------------+---------------------------------------+
3 rows in set (0.00 sec)
mysql>
mysql> SELECT * FROM support_ticket;
+-----------+-------------+----------+---------------------------+----------+---------------+--------------+
| ticket_id | customer_id | agent_id | issue_description | priority | ticket_status | created_date |
+-----------+-------------+----------+---------------------------+----------+---------------+--------------+
1	1	1	Product delivery delay	High	Open	2026-10-20
2	2	2	Login problem	Medium	In Progress	2026-10-21
3	3	3	Incorrect billing amount	High	Resolved	2026-10-22
+-----------+-------------+----------+---------------------------+----------+---------------+--------------+


MySQL 8.0 Command Line Cli
mysql> SELECT * FROM customer_interaction;
+----------------+-------------+----------+------------------+------------------+---+
| interaction_id | customer_id | agent_id | interaction_type | interaction_date | description |
+----------------+-------------+----------+------------------+------------------+---+
1	1	1	Phone Call	2026-10-20	Customer requested product information
2	2	2	Email	2026-10-21	Customer reported technical issue
3	3	3	Live Chat	2026-10-22	Customer asked about billing
+----------------+-------------+----------+------------------+------------------+---+
3 rows in set (0.00 sec)
mysql>
mysql> SELECT * FROM support_ticket;
+-----------+-------------+----------+--------------------------+----------+---------------+--------------+
| ticket_id | customer_id | agent_id | issue_description | priority | ticket_status | created_date |
+-----------+-------------+----------+--------------------------+----------+---------------+--------------+
1	1	1	Product delivery delay	High	Open	2026-10-20
2	2	2	Login problem	Medium	In Progress	2026-10-21
3	3	3	Incorrect billing amount	High	Resolved	2026-10-22
+-----------+-------------+----------+--------------------------+----------+---------------+--------------+
3 rows in set (0.00 sec)
mysql>
mysql> SELECT * FROM central_management;
+---------------+--------------+------------------+------------------+------------+
| management_id | manager_name | department | email | phone |
+---------------+--------------+------------------+------------------+------------+
| 1 | Ramesh | Operations | ramesh@gmail.com | 9876543216 |
| 2 | Divya | Customer Service | divya@gmail.com | 9876543217 |
+---------------+--------------+------------------+------------------+------------+
2 rows in set (0.00 sec)
mysql> UPDATE customer
    -> SET phone = '9999999999'
    -> WHERE customer_id = 1;
Query OK, 1 row affected (0.06 sec)
Rows matched: 1 Changed: 1 Warnings: 0
mysql> UPDATE support_ticket
    -> SET ticket_status = 'Resolved'
    -> WHERE ticket_id = 1;
Query OK, 1 row affected (0.06 sec)
Rows matched: 1 Changed: 1 Warnings: 0


MySQL 8.0 Command Line Cli
ysql> UPDATE support_ticket
   -> SET ticket_status = 'Resolved'
   -> WHERE ticket_id = 1;
uery OK, 1 row affected (0.06 sec)
ows matched: 1 Changed: 1 Warnings: 0
ysql> DELETE FROM customer
   -> WHERE customer_id = 3;
RROR 1451 (23000): Cannot delete or update a parent row: a foreign key constraint fails (`bpo_management`.`customer_interaction`, CONSTRAINT `custo
er_interaction_ibfk_1` FOREIGN KEY (`customer_id`) REFERENCES `customer` (`customer_id`))
ysql> DELETE FROM customer_interaction
   -> WHERE customer_id = 3;
uery OK, 1 row affected (0.01 sec)
ysql>
ysql> DELETE FROM support_ticket
   -> WHERE customer_id = 3;
uery OK, 1 row affected (0.00 sec)
ysql>
ysql> DELETE FROM customer
   -> WHERE customer_id = 3;
uery OK, 1 row affected (0.00 sec)
ysql> SELECT
   ->   c.customer_name,
   ->   a.agent_name,
   ->   i.interaction_type,
   ->   i.interaction_date,
   ->   i.description
   -> FROM customer_interaction i
   -> JOIN customer c
   -> ON i.customer_id = c.customer_id
   -> JOIN agent a
   -> ON i.agent_id = a.agent_id;
+---------------+------------+------------------+------------------+---+
| customer_name | agent_name | interaction_type | interaction_date | description |
+---------------+------------+------------------+------------------+---+
| Arun | Kiran | Phone Call | 2026-10-20 | Customer requested product information |
| Priya | Sneha | Email | 2026-10-21 | Customer reported technical issue |
+---------------+------------+------------------+------------------+---+
 rows in set (0.00 sec)


MySQL 8.0 Command Line Cli
-> i.description
-> FROM customer_interaction i
-> JOIN customer c
-> ON i.customer_id = c.customer_id
-> JOIN agent a
-> ON i.agent_id = a.agent_id;
customer_name	agent_name	interaction_type	interaction_date	description
Arun	Kiran	Phone Call	2026-10-20	Customer requested product information
Priya	Sneha	Email	2026-10-21	Customer reported technical issue
2 rows in set (0.00 sec)
mysql> SELECT
-> t.ticket_id,
-> c.customer_name,
-> a.agent_name,
-> t.issue_description,
-> t.priority,
-> t.ticket_status
-> FROM support_ticket t
-> JOIN customer c
-> ON t.customer_id = c.customer_id
-> JOIN agent a
-> ON t.agent_id = a.agent_id;
ticket_id	customer_name	agent_name	issue_description	priority	ticket_status
1	Arun	Kiran	Product delivery delay	High	Resolved
2	Priya	Sneha	Login problem	Medium	In Progress
2 rows in set (0.00 sec)
mysql> SELECT COUNT(*) AS total_customers
-> FROM customer;
| total_customers |
| 2 |
1 row in set (0.00 sec)


MySQL 8.0 Command Line Cli
-> t.issue_description,
-> t.priority,
-> t.ticket_status
-> FROM support_ticket t
-> JOIN customer c
-> ON t.customer_id = c.customer_id
-> JOIN agent a
-> ON t.agent_id = a.agent_id;
ticket_id	customer_name	agent_name	issue_description	priority	ticket_status
1	Arun	Kiran	Product delivery delay	High	Resolved
2	Priya	Sneha	Login problem	Medium	In Progress
2 rows in set (0.00 sec)
mysql> SELECT COUNT(*) AS total_customers
-> FROM customer;
| total_customers |
| 2 |
1 row in set (0.00 sec)
mysql> SELECT COUNT(*) AS pending_tickets
-> FROM support_ticket
-> WHERE ticket_status IN ('Open', 'In Progress');
| pending_tickets |
| 1 |
1 row in set (0.00 sec)
mysql>